

MINGYUE HUO 霍明月 /mɪŋjʊə huɔ/

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[Seeking Full-Time Research Scientist roles](#) | [Available Fall 2026](#)

EDUCATION

University of Illinois Urbana-Champaign

PhD in Computational Linguistics

MS in Applied Statistics

MA in Japanese Linguistics

USA

Dec 2026 (expected)

May 2026

May 2022

Shanghai International Studies University

BA in Japanese

China

June 2020

Hosei University (exchange)

2018, Tokyo, Japan

HIGHLIGHTS

Industry Research: Five research internships across Netflix, Tencent AI Lab, Amazon Prime Video, ByteDance AI Lab, and MERL, spanning speech, audio, and multimodal AI.

Research Focus: Speech and multimodal AI, with recent work focused on foundation-model evaluation and post-training, built on prior research in speech representation learning and multi-speaker understanding.

Publications: 7 first-author publications in leading speech and NLP venues, including ICASSP, Interspeech, ACL, ASRU, and the *Journal of the Acoustical Society of America*.

Open-source / Demos: end-to-end demos for [Speaker Diarization](#) (time-stamped meeting transcription for real-world application) and [Target Speaker Extraction](#). Contributor to open-source audio understanding toolkit [Auden](#).

PUBLICATIONS

SpeechCritic: Learning a Diagnostic Speech Judge from Limited Human Preferences

Mingyue Huo, Shivam Mehta, Bhavin Jawade, Yinghong Lan, Haoqi Li

In submission [\[PDF\]](#) [\[Demo\]](#)

Louder, Longer, Livelier: Acoustic Shortcuts and Misgrounded Rationales in Speech LLM Judges

Mingyue Huo, Shivam Mehta, Bhavin Jawade, Yinghong Lan, Haoqi Li

In submission [\[PDF\]](#) [\[Benchmark\]](#)

SESR: A Sound Event Structure Reward for Fine-grained Text-to-audio Alignment

Haoyu Zhang, Mingyue Huo, Dong Yang, Kiyoshi Kurihara, Irene Li, Yusuke Iwasawa, Yutaka Matsuo

In submission

Where Speech Enhancement Hurts Recognition: An Inference Time Polar Projection Diagnosis

Mingyue Huo, Yuheng Zhang, Hao Zhang

In submission [\[PDF\]](#)

Fooling Reward Models with Gibberish: Adversarial Examples Beyond the Semantic Regime

Yuheng Zhang*, Mingyue Huo*, Minghao Zhu, Mengxue Zhang, Nan Jiang (* Equal contribution)

In submission [\[PDF\]](#)

TagSpeech: End-to-End Multi-Speaker ASR and Diarization with Fine-Grained Temporal Grounding

Mingyue Huo, Yiwen Shao, Yuheng Zhang

ACL 2026 Main **Oral** [\[PDF\]](#) [\[Code+Demo\]](#) [\[Model\]](#)

Revisiting Audio-language Pretraining for Learning General-purpose Audio Representation

Wei-Cheng Tseng, Xuanru Zhou, Mingyue Huo, Yiwen Shao, Hao Zhang, Dong Yu

ACL 2026 Main [\[PDF\]](#)

Auden-Voice: General-Purpose Voice Encoder for Speech and Language Understanding

Mingyue Huo, Wei-Cheng Tseng, Yiwen Shao, Hao Zhang, Dong Yu

ICASSP 2026 [\[PDF\]](#) [\[Code\]](#) [\[Model\]](#)

Identifying and Calibrating Overconfidence in Noisy Speech Recognition

Mingyue Huo, Yuheng Zhang, Yan Tang

ASRU 2025 [\[PDF\]](#)

Beyond Speaker Identity: Text Guided Target Speech Extraction

Mingyue Huo, Abhinav Jain, Cong Phuoc Huynh, Fanjie Kong, Pichao Wang, Zhu Liu, Vimal Bhat

ICASSP 2025 [\[PDF\]](#) [\[Demo\]](#)

Iterative Nash Policy Optimization: Aligning LLMs with General Preferences via No-Regret Learning

Yuheng Zhang, Dian Yu, Baolin Peng, Linfeng Song, Ye Tian, **Mingyue Huo**, Nan Jiang, Haitao Mi, Dong Yu

ICLR 2025 **Oral** [\[PDF\]](#)

What makes listening comprehension difficult?: A feature-based machine learning approach to understanding item difficulty

Huiying Cai, Xun Yan, Ping-Lin Chuang, Yulin Pan, **Mingyue Huo**

Journal: Applied Linguistics, 2025 [\[PDF\]](#)

Release from Same-talker Speech-on-speech Masking: Effects of Masker Intelligibility and Other Contributing Factors

Mingyue Huo, Yinglun Sun, Dan Fogerty, Yan Tang

Journal: Journal of the Acoustical Society of America (JASA), 2024 [\[PDF\]](#)

Quantifying Informational Masking due to Masker Intelligibility in Same-talker Speech-in-speech Perception

Mingyue Huo, Yinglun Sun, Dan Fogerty, Yan Tang

INTERSPEECH 2023 **Oral** [\[PDF\]](#)

Towards Automating the Assessment of Teachers' Teaching Emotions in Preschool Classrooms

Mingyue Huo, Yan Tang

20th International Congress of Phonetic Sciences (ICPhS) 2023 [\[PDF\]](#)

INDUSTRY EXPERIENCE

Research Intern at MERL (Mitsubishi Electric Research Laboratories) Fall 2026, Cambridge, MA

Host: Jonathan Le Roux

- Long-form audio understanding in real-world multi-speaker meeting scenarios.

Research Scientist Intern at Netflix

Summer 2026, Los Gatos, CA

Host: Yinghong Lan

- Developed a diagnostic Audio LLM-as-a-Judge for cross-lingual generation and systematically evaluated Supervised Fine-Tuning (SFT), On-Policy Distillation (OPD), and Reinforcement Learning (RL) as post-training strategies.
- Designed a supervision amplification framework that uses calibrated acoustic signals as soft privileged information for Gemini, scaling scarce human supervision into large training sets.

Research Scientist Intern at Tencent AI-Lab

Summer 2025, Bellevue, WA

Host: Dong Yu

- Developed general-purpose speech representations by systematically adapting and benchmarking representation-learning methods from vision for speech, evaluating learned representations across speaker identity, paralinguistics, speech-text retrieval, and audio-language understanding.
- Built general-purpose speech encoders for speaker verification, diarization, speaker counting, and LLM-based audio QA; contributed to the open-source Auden toolkit, with resulting research accepted at ICASSP 2026.

Applied Scientist Intern at Amazon Prime Video

Summer 2024, Seattle, WA

Host: Zhu Liu

- Developed a text-guided target speech extraction model that conditions on natural-language descriptions to identify and extract target speakers without traditional audio identifiers.
- Built a speech–mixture–text–description dataset and evaluation pipeline, achieving an average 16 dB SI-SDR improvement; resulting research was accepted at ICASSP 2025.

Linguist Intern at ByteDance AI-Lab

Dec 2019 – July 2021, Shanghai, China

- Worked on Mandarin and Japanese TTS and singing synthesis, designing annotation pipelines, grapheme-to-phoneme conversion rules, and linguistically targeted evaluation sets for model development.
- Evaluated generative speech models using expert analysis, MOS, and A/B testing to improve naturalness and expressiveness; several resulting voices were launched on TikTok.

AWARDS

- UIUC Graduate College Dissertation Completion Fellowship for 2026-2027
- Conference Travel Award – UIUC Linguistics November 2025
- C.W. Kim Research Award in Linguistics – UIUC May 2024
- Linguistics Student Leadership Award – UIUC May 2024
- Midwest Speech and Language Days Travel Award – University of Michigan April 2024
- Departmental Conference Travel Award – UIUC November 2023
- Best Japanese TA – UIUC May 2022
- EALC Summer Research Fellowship – UIUC May 2021
- Teachers Ranked as Excellent by Their Students – UIUC 2021 Fall and 2020 Fall

COMMUNITY SERVICE

- **Student Volunteer** at INTERSPEECH 2021, NeurIPS 2023
- **Reviewer** at ICASSP 2026, INTERSPEECH 2025 & 2026, SLT2026

SKILLS

- **Language:** Mandarin (Native), English (Advanced), Japanese (Advanced N1)
- **Software:** Praat, Audacity, Adobe Audition, Adobe Premiere, Montreal Forced Aligner
- **Programming:** Python, TensorFlow, MATLAB, Shell, R, $\LaTeX$
- **Framework & Library:** PyTorch, Kaldi, NumPy, SciPy, TorchAudio, librosa, soundfile, Matplotlib, Lhotse
- **Highlighted Coursework:** Digital Signal Processing (A+), Psychoacoustics (A), Phonetics (A+), Phonology (A), Formal Semantics (A+), Syntax (A), Machine Learning (A), Natural Language Processing (A), Generative Models for Audio (A)

Last Updated on Sep.20 2026